WORKED SOLUTION

STEP I 1997, Question 5

Pure • Vectors in Geometry

STEP I 1997, Question 5

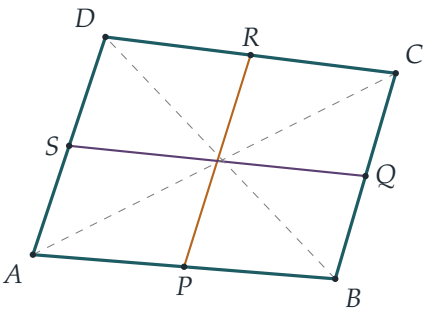
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q5

The key idea

Lengths squared are dot products, and mid-points are averages, so position vectors turn the whole identity into algebra. The useful trick is to spot that both sides secretly involve the diagonals AC and BD: once each difference is written in terms of $\overrightarrow{AC} \cdot \overrightarrow{BD}$, the identity falls out. The deduction then only needs the fact that the four rod lengths never change.

Solution



Let A, B, C, D have position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ relative to any origin, and write $|\mathbf{x}|^2 = \mathbf{x} \cdot \mathbf{x}$. The mid-points are

$$\mathbf{p} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}), \quad \mathbf{q} = \tfrac{1}{2}(\mathbf{b}+\mathbf{c}), \quad \mathbf{r} = \tfrac{1}{2}(\mathbf{c}+\mathbf{d}), \quad \mathbf{s} = \tfrac{1}{2}(\mathbf{d}+\mathbf{a}).$$

The four sides

Expanding each $|\mathbf{y}-\mathbf{x}|^2 = |\mathbf{x}|^2 + |\mathbf{y}|^2 - 2\,\mathbf{x}\cdot\mathbf{y}$, the squared terms $|\mathbf{a}|^2, \ldots, |\mathbf{d}|^2$ each appear once on each side and cancel, leaving

$$\begin{aligned} |AB|^2 + |CD|^2 - |AD|^2 - |BC|^2 &= -2\,\mathbf{a}\cdot\mathbf{b} - 2\,\mathbf{c}\cdot\mathbf{d} + 2\,\mathbf{a}\cdot\mathbf{d} + 2\,\mathbf{b}\cdot\mathbf{c} \\ &= 2(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b}). \end{aligned}$$

(Multiply out $(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b})$ to confirm the factorisation.) In words, this is $-2\,\overrightarrow{AC}\cdot\overrightarrow{BD}$.

The two mid-point joins

We have

$$\overrightarrow{PR} = \tfrac{1}{2}(\mathbf{c}+\mathbf{d}-\mathbf{a}-\mathbf{b}) = \tfrac{1}{2}\mathbf{u}, \qquad \overrightarrow{QS} = \tfrac{1}{2}(\mathbf{d}+\mathbf{a}-\mathbf{b}-\mathbf{c}) = \tfrac{1}{2}\mathbf{v},$$

say.

> **Ask yourself** How can we find $|\mathbf{u}|^2 - |\mathbf{v}|^2$ without expanding two four-term squares?

Use the difference of two squares for vectors: $|\mathbf{u}|^2 - |\mathbf{v}|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}-\mathbf{v})$, which holds because $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$. Here

$$\mathbf{u}+\mathbf{v} = 2(\mathbf{d}-\mathbf{b}), \qquad \mathbf{u}-\mathbf{v} = 2(\mathbf{c}-\mathbf{a}),$$

so

$$|PR|^2 - |QS|^2 = \tfrac{1}{4}(|\mathbf{u}|^2 - |\mathbf{v}|^2) = (\mathbf{c}-\mathbf{a})\cdot(\mathbf{d}-\mathbf{b}) = \overrightarrow{AC}\cdot\overrightarrow{BD}.$$

Putting them together

Adding the side result to twice the mid-point result,

$$\begin{aligned} &|AB|^2 + |CD|^2 - |AD|^2 - |BC|^2 + 2(|PR|^2 - |QS|^2) \\ &\quad = 2(\mathbf{a}-\mathbf{c})\cdot(\mathbf{d}-\mathbf{b}) + 2(\mathbf{c}-\mathbf{a})\cdot(\mathbf{d}-\mathbf{b}) = 0, \end{aligned}$$

which rearranges to

$$|AB|^2 + |CD|^2 + 2|PR|^2 = |AD|^2 + |BC|^2 + 2|QS|^2,$$

as required. We never used convexity, or even that the points lie in a plane, so the identity holds for any four points, including a quadrilateral that has folded over so that two of its rods cross.

The deduction

Rearranging the identity,

$$|PR|^2 - |QS|^2 = \tfrac{1}{2}(|AD|^2 + |BC|^2 - |AB|^2 - |CD|^2).$$

The rods are rigid, so the four lengths on the right never change, however the joints flex. So $|PR|^2 - |QS|^2$ is constant throughout the motion. Equivalently, the scalar product $\overrightarrow{AC} \cdot \overrightarrow{BD}$ of the diagonals stays fixed, even though the diagonals themselves change length and direction.

Checks

- For a rectangle with sides a and b, $|AB| = |CD| = a$ and $|AD| = |BC| = b$, while PR and QS join mid-points of opposite sides, so $|PR| = b$ and $|QS| = a$. Both sides of the identity equal $2a^2 + 2b^2$.
- For a rhombus all four sides are equal, so the deduction says $|PR| = |QS|$ in every position. Indeed, the diagonals of a rhombus are perpendicular, so $\overrightarrow{AC} \cdot \overrightarrow{BD} = 0$.
- Numerically, the identity holds to within 10^{-12} on 10^5 random quadrilaterals. For a linkage with $|AB| = 3$, $|BC| = 4$, $|CD| = 5$, $|DA| = 6$ flexed through about 200 positions, $|PR|^2 - |QS|^2$ stayed at $\frac{1}{2}(36 + 16 - 9 - 25) = 9$ throughout.

Common slips

- Mixing up which sides pair with which: PR joins the mid-points of AB and CD, and it is those two sides that sit with PR in the identity.
- Using a coordinate set-up such as A at the origin and B on an axis, then forgetting that the general quadrilateral needs every other point free. Vectors avoid this bookkeeping.

Beyond STEP

The points P, Q, R, S always form a parallelogram (Varignon's theorem), and PR, QS are its diagonals. The identity here is a cousin of the parallelogram law $|\mathbf{u} + \mathbf{v}|^2 + |\mathbf{u} - \mathbf{v}|^2 = 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2$, which in a first-year linear algebra course characterises exactly those norms that come from an inner product. The idea of a quantity that stays fixed while a mechanism moves, an invariant, recurs throughout mathematics. Strang's *Introduction to Linear Algebra* is a good place to see dot products and inner products developed properly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.